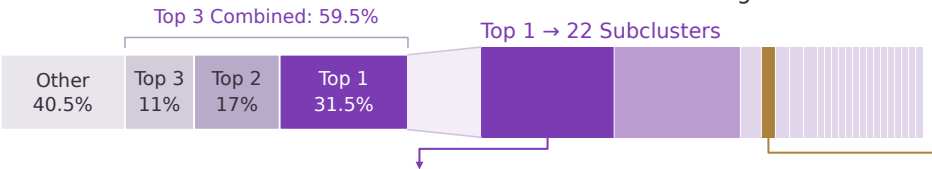


Pattern Distribution



Question A

A sequence of numbers is defined by $a_1 = 2$, $a_2 = 5$, and for all $n \geq 3$, a_n is the smallest positive integer greater than a_{n-1} that is relatively prime to a_{n-2} . What is the remainder when the sum of the first 20 terms of this sequence is divided by 17?

Question B

The sequence of numbers $b_0, b_1, b_2, \dots$ is defined as follows: $b_0 = 2$, $b_1 = 5$, and for $n \geq 2$, b_n is the smallest positive integer greater than b_{n-1} that is relatively prime to b_{n-2} . What is the remainder when the sum $b_0 + b_1 + \dots + b_{19}$ is divided by 17?

Note: Two numbers are relatively prime if their greatest common divisor (GCD) is 1.

Question A and B are equivalent mathematical tasks

The same sequence, indexed from 1 in A and 0 in B. Both ask for the remainder when the sum of the first 20 terms is divided by 17.